



US012408756B1

(12) **United States Patent**
Olarte

(10) **Patent No.:** **US 12,408,756 B1**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **STACKING CHAIR WITH REMOVABLE BACK**

(71) Applicant: **Series International, LLC**, Aventura, FL (US)

(72) Inventor: **Alvaro Mauricio Olarte**, Aventura, FL (US)

(73) Assignee: **Series International, LLC**, Aventura, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 115 days.

(21) Appl. No.: **18/325,463**

(22) Filed: **May 30, 2023**

Related U.S. Application Data

(60) Provisional application No. 63/346,638, filed on May 27, 2022.

(51) **Int. Cl.**
A47C 3/04 (2006.01)
A47C 4/02 (2006.01)
A47C 5/06 (2006.01)
A47C 5/10 (2006.01)
A47C 7/02 (2006.01)
A47C 7/18 (2006.01)
A47C 7/22 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC *A47C 3/04* (2013.01); *A47C 4/02* (2013.01); *A47C 4/028* (2013.01); *A47C 5/06* (2013.01); *A47C 5/10* (2013.01); *A47C 7/0213* (2018.08); *A47C 7/18* (2013.01); *A47C 7/22* (2013.01); *A47C 7/24* (2013.01); *A47C 7/42* (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

62,428 A 2/1867 Koechling
146,432 A 1/1874 Close
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2008219128 A1 8/2008
AU 2019204076 B2 1/2020
(Continued)

OTHER PUBLICATIONS

Canadian Office Action, PCT US2019061755, Jul. 27, 2022.
IP Australia, Australian Government Patent Examination Report No. 1 Application No. 2012304735, Jun. 18, 2015.

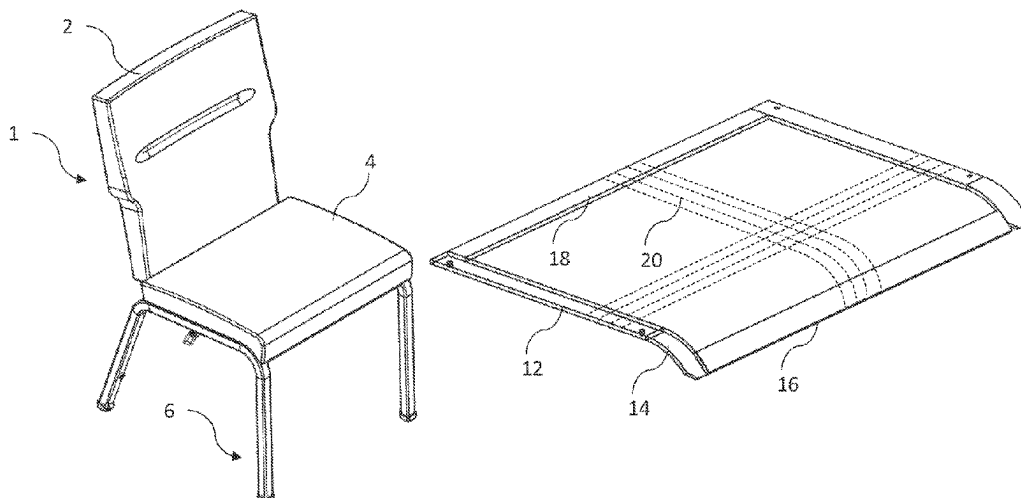
Primary Examiner — Robert Canfield

(74) *Attorney, Agent, or Firm* — MALLOY & MALLOY, PL.

(57) **ABSTRACT**

A stacking chair system comprising a frame with four legs, two being front legs and two being rear legs. The seat is secured to the frame, the seat being upholstered and including a foam cushion supported by flexible support. Two receivers in the frame are at a rear of the frame and between the rear legs of the frame. A backrest comprising a foam cushion is upholstered and has an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. A fastener is provided for each receiver and is configured to secure the protrusion to the corresponding receiver. The backrest is removable upon removal of the fasteners.

18 Claims, 6 Drawing Sheets



(51)	Int. Cl.		3,446,530 A	5/1969	Rowland
	<i>A47C 7/24</i>		3,479,084 A	11/1969	Van Ryn
	<i>A47C 7/42</i>		D216,305 S	12/1969	Gazit
	(2006.01)		3,506,167 A	4/1970	Orr
	(2006.01)		3,528,096 A	9/1970	Moberg
			3,531,552 A	9/1970	Getz et al.
(56)	References Cited		3,544,163 A	12/1970	Krein
	U.S. PATENT DOCUMENTS		3,547,488 A	12/1970	Barnes
			3,550,958 A	12/1970	Krein
	216,307 A	6/1879 Case	3,556,588 A	1/1971	Monyer et al.
	252,533 A	1/1882 Smith	3,583,590 A	6/1971	Ferraro
	272,166 A	2/1883 Shepler	3,589,762 A	6/1971	Henrikson
	315,618 A	4/1885 Harwood	3,594,037 A	7/1971	Sherman
	326,571 A	9/1885 Lancaster	D221,642 S	8/1971	Bayes
	646,290 A	3/1900 Kratz	D222,350 S	10/1971	Sundberg
	669,089 A	3/1901 Lambert	3,641,614 A	2/1972	Newsome
	688,878 A	12/1901 Neuendorff	3,647,260 A	3/1972	Grant et al.
	1,010,557 A	12/1911 Wilkinson	3,657,854 A	4/1972	Tipton
	1,152,480 A	9/1915 Bouk	3,695,687 A	10/1972	Uyeda
	1,374,467 A	4/1921 Park	3,695,693 A	10/1972	Acton et al.
	1,662,378 A	3/1928 Duke	3,714,678 A	2/1973	Weisz et al.
	1,695,101 A	12/1928 Hoffman	3,717,289 A	2/1973	Laurizio
	1,743,802 A	1/1930 Andress	3,722,950 A	3/1973	Harnick
	1,800,387 A	4/1931 Greist	3,727,975 A	4/1973	Anderson
	1,810,888 A	6/1931 Hanson	3,744,843 A	7/1973	Barecki et al.
	1,922,582 A	8/1933 Goodrich	D228,104 S	8/1973	Blodee
	1,941,340 A	12/1933 Dellert	3,757,984 A	9/1973	Barton
	1,973,178 A	9/1934 Sass	3,759,572 A	9/1973	Koepke
	2,000,172 A	5/1935 Hanson	3,762,765 A	10/1973	Piretti
	2,044,473 A	6/1936 Geller	3,762,766 A	10/1973	Barecki et al.
	2,070,387 A	2/1937 Vandervoort	3,784,249 A	1/1974	Hendrickson et al.
	2,098,711 A	11/1937 Rastetter	3,785,600 A	1/1974	Padovano
	2,232,205 A	2/1941 Boardman	3,796,459 A	3/1974	Weber
	2,290,556 A	7/1942 Hard	3,813,149 A	5/1974	Lawrence, III et al.
	2,304,223 A	12/1942 Westrope	3,847,433 A	11/1974	Acton et al.
	2,492,103 A	12/1949 Merrill	3,850,476 A	11/1974	Day
	2,492,111 A	12/1949 Prosser et al.	3,857,605 A	12/1974	Fantoni
	D156,935 S	1/1950 Carlson	3,858,766 A	1/1975	Schiemann
	2,500,124 A	3/1950 Hoven	3,874,731 A	4/1975	Jordan
	2,516,728 A	7/1950 Smith	3,877,674 A	4/1975	Cerutti
	2,518,381 A	8/1950 Runkles	3,880,467 A	4/1975	Tischler
	D165,457 S	12/1951 Hoven et al.	3,885,766 A	5/1975	Resch et al.
	2,608,238 A	8/1952 Larsen et al.	3,889,999 A	6/1975	Mackintosh
	2,610,335 A	9/1952 Baptista	3,899,207 A	8/1975	Mueller
	2,610,669 A	9/1952 Bromagem	3,901,417 A	8/1975	Schiemann
	2,620,858 A	12/1952 Hoven et al.	3,902,754 A	9/1975	Braeuning
	2,621,709 A	12/1952 Bell	3,933,268 A	1/1976	Buske
	2,652,881 A	9/1953 Rowe	3,960,405 A	6/1976	DeLong
	2,688,997 A	9/1954 Miller	3,976,326 A	8/1976	Grumblatt
	2,750,993 A	6/1956 McGregor	4,000,586 A	1/1977	Vance et al.
	2,812,120 A	11/1957 Beall, Jr.	4,052,101 A	10/1977	DeLong
	2,815,972 A	12/1957 Lagervall	4,054,316 A	10/1977	DeLong
	2,828,803 A	4/1958 Howe et al.	4,077,669 A	3/1978	Fox
	2,842,187 A	7/1958 Hendrickson	4,097,089 A	6/1978	Petersen
	2,853,119 A	9/1958 Balfour	4,114,947 A	9/1978	Nelson
	2,913,039 A	11/1959 Mauser	4,126,354 A	11/1978	DeLong et al.
	2,915,223 A	12/1959 Beall, Jr.	4,133,579 A	1/1979	Springfield
	2,942,652 A	6/1960 Chapman	D253,034 S	10/1979	Meyer et al.
	2,986,201 A	5/1961 McCartan	4,179,158 A	12/1979	Flaum et al.
	3,087,755 A	4/1963 Boman	4,189,876 A	2/1980	Crossman et al.
	3,098,677 A	7/1963 Williams	4,216,994 A	8/1980	Benoit
	3,102,754 A	9/1963 Junkunc	4,240,663 A	12/1980	Locher
	3,114,575 A	12/1963 Eames	D258,866 S	4/1981	Houseman et al.
	3,137,527 A	6/1964 Hoven et al.	4,273,265 A	6/1981	Anderson
	3,170,731 A	2/1965 Caldemeyer et al.	4,277,101 A	7/1981	Vogel
	3,173,587 A	3/1965 Stearns	D260,058 S	8/1981	Ponzellini
	3,182,377 A	5/1965 Hoven et al.	4,297,763 A	11/1981	Lautenschlager
	3,194,601 A	7/1965 Hoven et al.	4,308,961 A	1/1982	Kunce
	3,197,254 A	7/1965 Hendrickson	4,330,898 A	5/1982	Thompson et al.
	3,233,939 A	2/1966 Chapman	D266,017 S	8/1982	Kellogg
	3,246,927 A	4/1966 Klassen	4,348,053 A	9/1982	Strassle
	3,264,034 A	8/1966 Lawson	4,453,363 A	6/1984	Koller
	3,300,246 A	1/1967 Bouche	D274,699 S	7/1984	Epperson
	3,305,044 A	2/1967 Van Loo et al.	4,493,172 A	1/1985	Jones
	3,310,300 A	3/1967 Lawson	4,509,665 A	4/1985	Goodall
	D208,468 S	9/1967 Okiyama	4,528,786 A	7/1985	Dinitz et al.
	3,343,870 A	9/1967 Thatcher et al.	4,542,887 A	9/1985	Bethell et al.
	3,369,658 A	2/1968 Hasselmann	4,547,092 A	10/1985	Vetter et al.
	D210,364 S	3/1968 Lindberg	4,575,149 A	3/1986	Forestal et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,603,907	A	8/1986	Witzke	5,549,072	A	8/1996	Maloney	
4,632,457	A	12/1986	Hofrichter et al.	5,553,923	A	9/1996	Bilezikjian	
4,650,100	A	3/1987	Echazabal, Jr.	5,567,016	A	10/1996	Koprowski	
4,691,828	A	9/1987	Slusarczyk et al.	5,597,097	A	1/1997	Morris	
4,694,531	A	9/1987	Foy	5,598,785	A	2/1997	Zaguroli, Jr.	
4,702,522	A	10/1987	Vail et al.	5,601,335	A	2/1997	Woods et al.	
4,705,274	A	11/1987	Lubeck	5,609,395	A	3/1997	Burch	
4,723,816	A	2/1988	Selbert et al.	5,632,406	A	5/1997	Robbins, III	
4,725,027	A	2/1988	Bekanich	5,636,900	A	6/1997	Wilkie et al.	
4,726,554	A	2/1988	Sorrell	5,651,908	A	7/1997	Mansfield	
4,748,913	A	6/1988	Favaretto et al.	5,658,043	A	8/1997	Davidson	
4,754,943	A	7/1988	Froutzis	5,661,945	A	9/1997	Henriksson et al.	
4,761,035	A	8/1988	Urai	5,671,868	A	9/1997	Herr	
4,766,838	A	8/1988	Johnson	5,671,975	A	9/1997	Muller	
4,789,126	A	12/1988	Rice et al.	5,681,085	A	10/1997	Nahoul	
4,790,594	A	12/1988	Temos	5,681,089	A	10/1997	Lin	
4,799,632	A	1/1989	Baymak et al.	5,683,136	A	11/1997	Baumann et al.	
4,805,793	A	2/1989	Brandt et al.	5,702,157	A	12/1997	Hurite	
4,828,208	A	5/1989	Peterson et al.	5,704,691	A	1/1998	Olson	
4,850,159	A	7/1989	Conner	5,711,355	A	1/1998	Kowalczyk	
4,850,502	A	7/1989	Davis	5,733,010	A	3/1998	Lewis et al.	
4,852,940	A	8/1989	Kanigowski	D393,803	S	4/1998	Walter	
4,854,640	A	8/1989	Yokoyama	5,740,947	A	4/1998	Flaig et al.	
4,858,996	A	8/1989	Blodee	5,746,358	A	5/1998	Crosby	
4,864,690	A	9/1989	Chen	5,762,296	A	6/1998	Gilbert	
4,865,377	A	9/1989	Musser et al.	5,762,396	A *	6/1998	Barile	A47C 3/04 297/239
4,883,319	A	11/1989	Scott	5,765,911	A	6/1998	Sorenson	
4,900,085	A	2/1990	Tobler	D396,967	S	8/1998	Yamazaki	
4,955,575	A	9/1990	Moore	5,788,332	A	8/1998	Hettinga	
4,960,137	A	10/1990	Pott et al.	5,803,546	A	9/1998	Yamazaki	
4,984,768	A	1/1991	Kolber et al.	5,826,850	A	10/1998	Goldsmith	
4,989,915	A	2/1991	Hansal	5,845,964	A	12/1998	Phoon	
5,002,337	A	3/1991	Engel et al.	5,850,949	A	12/1998	Koerbel et al.	
D315,867	S	4/1991	Hestehave et al.	5,878,470	A	3/1999	Blansett	
5,020,286	A	6/1991	Hamar et al.	5,881,431	A	3/1999	Pieper, II et al.	
5,033,792	A	7/1991	Kanazawa	5,890,761	A	4/1999	Miller	
5,067,772	A	11/1991	Koa	5,897,035	A	4/1999	Schlomer	
5,087,096	A	2/1992	Yamazaki	5,899,526	A	5/1999	LaPointe et al.	
5,098,156	A	3/1992	Vogel	5,899,531	A	5/1999	Koehler	
5,108,016	A	4/1992	Waring	5,924,608	A	7/1999	Chiu	
5,110,013	A	5/1992	Clark et al.	5,931,528	A	8/1999	Shields	
5,121,891	A	6/1992	Goldsmith	5,944,387	A	8/1999	Stumpf	
5,123,702	A	6/1992	Caruso	5,944,388	A	8/1999	Saucier et al.	
5,142,757	A	9/1992	Thary	5,971,466	A	10/1999	Hashimoto	
D331,703	S	12/1992	Seguin et al.	5,984,417	A	11/1999	Wang	
5,167,336	A	12/1992	Lajovic	5,985,188	A	11/1999	Jennings et al.	
D335,453	S	5/1993	Marti	5,988,757	A	11/1999	Vishey et al.	
5,232,110	A	8/1993	Purnell	6,012,773	A	1/2000	Best	
5,236,165	A	8/1993	Conway	6,019,413	A	2/2000	Scraver et al.	
5,236,247	A	8/1993	Hewko	6,024,397	A	2/2000	Scraver et al.	
5,246,270	A	9/1993	Vogel	6,029,858	A	2/2000	Srokose et al.	
5,282,662	A	2/1994	Bolsworth et al.	6,045,013	A	4/2000	Yang	
5,288,128	A	2/1994	Smith	6,050,455	A	4/2000	Soehnlén et al.	
5,306,072	A	4/1994	Caldwell	6,056,278	A	5/2000	Bullard et al.	
5,308,126	A	5/1994	Weger, Jr. et al.	6,068,161	A	5/2000	Soehnlén et al.	
5,316,375	A	5/1994	Breen	D426,162	S	6/2000	Olson	
5,328,231	A	7/1994	Raymond	6,073,997	A	6/2000	Koh	
5,329,871	A	7/1994	Gibbs	6,082,591	A	7/2000	Healey	
5,342,110	A	8/1994	Merrick	6,092,864	A	7/2000	Wycech et al.	
5,375,914	A	12/1994	Donnelly	6,095,603	A	8/2000	Hock	
5,377,882	A	1/1995	Pham et al.	6,109,696	A *	8/2000	Newhouse	A47C 7/42 297/452.21
5,383,644	A	1/1995	Huse	6,116,183	A	9/2000	Crow et al.	
5,393,126	A	2/1995	Boulva	6,135,562	A	10/2000	Infanti	
5,400,928	A	3/1995	Resnick	6,145,912	A	11/2000	Rice et al.	
5,406,994	A	4/1995	Mitchell et al.	6,161,262	A	12/2000	Pfister	
D358,330	S	5/1995	Kahl	6,174,029	B1 *	1/2001	Swy	A47C 3/04 297/440.22
5,433,346	A	7/1995	Howe	D438,119	S	2/2001	Meyer et al.	
5,470,128	A	11/1995	Kerkham	6,190,047	B1	2/2001	Haag et al.	
5,472,124	A	12/1995	Martushev	6,196,425	B1	3/2001	Fielding et al.	
D365,983	S	1/1996	Triassi et al.	6,206,469	B1	3/2001	Caruso et al.	
5,524,963	A *	6/1996	Barile	6,220,658	B1	4/2001	Lukawski et al.	
				6,220,661	B1	4/2001	Peterson	
				D441,972	S	5/2001	Mitjans	
5,529,376	A	6/1996	Jovan et al.	6,224,149	B1	5/2001	Gevaert	
5,538,165	A	7/1996	Frohn	6,226,819	B1	5/2001	Ogawa et al.	
5,547,247	A	8/1996	Dixon	6,230,944	B1	5/2001	Castellano et al.	

(56)

References Cited

U.S. PATENT DOCUMENTS

D445,269	S	7/2001	Mitjans	7,108,447	B2	9/2006	Akkala et al.
D445,574	S	7/2001	Foster	D530,936	S	10/2006	Gehry
D446,035	S	8/2001	Haney	7,134,728	B1	11/2006	Buhrman
D447,642	S	9/2001	Kimura	7,150,498	B2	12/2006	Svartvatn
6,283,550	B1	9/2001	Vialatte et al.	7,182,214	B2	2/2007	Darr et al.
6,286,902	B1	9/2001	Yoshimura	D537,648	S	3/2007	Moschini
6,293,621	B1	9/2001	Marsh et al.	7,204,553	B2	4/2007	Olarte
6,296,222	B1	10/2001	Whittaker	7,210,735	B2	5/2007	Lang
6,296,315	B1	10/2001	Jensen	7,225,899	B2	6/2007	Molnar et al.
D450,245	S	11/2001	Schultz	D549,983	S	9/2007	Olarte
6,314,614	B1	11/2001	Kuehl	D549,984	S	9/2007	Olarte
6,345,863	B1	2/2002	Laws et al.	D556,058	S	11/2007	Bowers et al.
6,349,449	B1	2/2002	Kuehl	7,303,235	B1	12/2007	Fongers
6,352,307	B1	3/2002	Engman	D558,604	S	1/2008	Dohm et al.
6,360,924	B1	3/2002	Franzen	D559,108	S	1/2008	Regas
6,382,475	B1	5/2002	Weikinger	D560,921	S	2/2008	Olarte
6,382,726	B2	5/2002	Bullesbach et al.	7,331,490	B2	2/2008	Yamana
6,393,624	B1	5/2002	Iwashita	7,370,910	B2	5/2008	Piretti
D458,140	S	6/2002	Hutchins	7,384,102	B2	6/2008	Chen et al.
6,409,267	B1	6/2002	Pietrzak	D573,473	S	7/2008	Kruparova
6,425,153	B1	7/2002	Reswick	D577,589	S	9/2008	Briessell et al.
6,427,287	B1	8/2002	Brueckner et al.	7,467,453	B2	12/2008	Olarte
6,428,100	B1	8/2002	Kain et al.	7,469,966	B1	12/2008	Vallee
6,450,579	B1	9/2002	Nylander et al.	7,478,876	B2	1/2009	Olarte
D465,664	S	11/2002	Stannard	7,490,392	B2	2/2009	Peterson
6,474,744	B1	11/2002	Taylor et al.	D589,807	S	4/2009	Gundrum et al.
6,477,738	B1	11/2002	Kremer et al.	D589,809	S	4/2009	Gundrum et al.
6,478,381	B1	11/2002	Cramb, III et al.	7,513,394	B2	4/2009	Bone
6,491,346	B1	12/2002	Gupta et al.	D594,669	S	6/2009	Asano
6,494,344	B1	12/2002	Kressel, Sr.	7,562,937	B2	7/2009	Baumann
6,499,158	B1	12/2002	Easterling	7,594,696	B2	9/2009	Girard
6,523,900	B1	2/2003	Conner et al.	7,661,549	B2	2/2010	Rae
6,554,353	B1	4/2003	Yu	7,690,732	B2	4/2010	Olarte
6,565,157	B2	5/2003	Barile, Jr. et al.	7,695,061	B2	4/2010	Olarte
6,578,922	B2	6/2003	Khedira et al.	D616,658	S	6/2010	Niermeyer
6,582,020	B1	6/2003	Tenenboym et al.	7,774,880	B1	8/2010	Botts
6,588,612	B1	7/2003	Dorn et al.	D625,774	S	10/2010	Nirenberg
D479,468	S	9/2003	Dorn et al.	7,814,636	B2	10/2010	Olarte
D479,657	S	9/2003	Calatrava Valls	7,828,380	B2	11/2010	Olarte
6,612,652	B1	9/2003	Tenenboym et al.	7,866,689	B2	1/2011	Sabaran
6,669,282	B2	12/2003	Piretti	7,882,795	B1	2/2011	Snyder
D485,170	S	1/2004	Hay et al.	7,901,004	B2	3/2011	Figueras Mitjans
6,683,394	B1	1/2004	Gevaert	7,905,546	B2	3/2011	Conner
6,698,834	B2	3/2004	Olarte	7,950,739	B2	5/2011	King
6,729,685	B1	5/2004	Ebalobor	7,959,044	B1	6/2011	Christian et al.
6,746,086	B1	6/2004	Foster	7,971,935	B2	7/2011	Saez et al.
6,786,548	B2	9/2004	Pearce et al.	8,020,254	B2	9/2011	Lin
6,786,549	B1	9/2004	Olarte	8,075,055	B2	12/2011	Olarte
6,789,549	B2	9/2004	Johnson	8,091,744	B2	1/2012	Zwahlen
6,793,281	B2	9/2004	Duerr et al.	8,091,945	B2	1/2012	Hancock et al.
6,805,394	B2	10/2004	Kettler et al.	8,104,838	B2	1/2012	Tsai
D498,077	S	11/2004	Tonizzo	8,109,566	B2	2/2012	Koh
6,839,950	B2	1/2005	Guillot	8,157,132	B1	4/2012	Johnson et al.
6,845,885	B2	1/2005	Morgenroth	8,201,588	B2	6/2012	Bonner
6,857,699	B2	2/2005	Ashton et al.	8,201,699	B2	6/2012	Zummo et al.
6,874,850	B2	4/2005	Berkowicz	8,245,891	B2	8/2012	Eriksen
D506,137	S	6/2005	Durivage	D672,657	S	12/2012	Hall
6,907,703	B2	6/2005	Gonzalez	D672,980	S	12/2012	Mitjans
6,932,228	B1	8/2005	Darr et al.	8,414,072	B2	4/2013	Phillips
D509,144	S	9/2005	Snyder	8,490,335	B2	7/2013	LaForest
6,942,295	B1	9/2005	Lopez	8,491,053	B2	7/2013	Stringer
6,964,345	B2	11/2005	Wetherell, Jr. et al.	D687,642	S	8/2013	Hsuan-Chin
6,968,983	B2	11/2005	Laible	D687,921	S	8/2013	Jung et al.
D514,940	S	2/2006	Darr et al.	D687,922	S	8/2013	Jung et al.
7,000,794	B2	2/2006	Soehnlen et al.	D689,297	S	9/2013	Su
7,000,989	B2	2/2006	Fisher	D690,203	S	9/2013	Hall et al.
D517,413	S	3/2006	Murphy	D690,952	S	10/2013	Becker
D519,378	S	4/2006	Abadinsky	D693,426	S	11/2013	Mehra et al.
7,032,272	B2	4/2006	Haenlein	8,573,704	B2	11/2013	Peters et al.
D520,367	S	5/2006	Morris	8,578,974	B2	11/2013	Bonner
D520,765	S	5/2006	Piano	D696,956	S	1/2014	Jung et al.
7,063,381	B2	6/2006	Scahill et al.	8,640,930	B2	2/2014	Nunez et al.
7,073,858	B2	7/2006	Fisher et al.	8,662,322	B2	3/2014	Magnusson et al.
D528,425	S	9/2006	Van Dorin et al.	8,662,359	B1	3/2014	Hickey
7,100,983	B1	9/2006	Gant	D703,458	S	4/2014	Nakamura et al.
				8,721,003	B2	5/2014	Waite et al.
				8,727,187	B2	5/2014	Magley
				8,727,445	B2	5/2014	De Maina
				8,763,826	B1	7/2014	Smith et al.

(56)	References Cited			2001/0054628 A1	12/2001	Harbaugh et al.	
	U.S. PATENT DOCUMENTS			2002/0038970 A1	4/2002	Leng et al.	
				2002/0053822 A1 *	5/2002	Ware	A47C 7/445
							297/239
	8,813,657 B2	8/2014	Winter et al.	2002/0117885 A1 *	8/2002	Barile, Jr.	A47C 7/282
	8,820,827 B2	9/2014	Olarte				297/452.52
	8,820,836 B2	9/2014	Stewart et al.	2002/0166857 A1	11/2002	Schwaikert	
	8,827,369 B2	9/2014	Izawa et al.	2002/0167789 A1	11/2002	Novin et al.	
	8,833,617 B1	9/2014	Compton et al.	2003/0047976 A1	3/2003	Hannon et al.	
	8,910,835 B2	12/2014	Ouderkirk	2003/0062759 A1	4/2003	Gupta et al.	
	8,960,454 B2	2/2015	Magnusson et al.	2003/0085607 A1	5/2003	Jones et al.	
	8,967,714 B2	3/2015	Machael et al.	2003/0090136 A1	5/2003	Olarte	
	8,968,041 B2	3/2015	Rubbo	2003/0102703 A1	6/2003	Tenenboym et al.	
	8,973,990 B2	3/2015	Krupiczewicz et al.	2003/0132184 A1	7/2003	Dorn et al.	
	9,033,414 B2	5/2015	Olarte	2004/0041455 A1	3/2004	Tenbroeck et al.	
	9,060,614 B1	6/2015	Gibilterra	2004/0084943 A1	5/2004	Fisher	
	D734,621 S	7/2015	Su	2004/0099672 A1	5/2004	Perlman	
	9,096,357 B2	8/2015	Brausen	2004/0100134 A1	5/2004	Plant et al.	
	9,113,715 B2	8/2015	Olarte	2004/0178222 A1	9/2004	Clausen et al.	
	9,295,334 B2	3/2016	Olarte	2004/0217635 A1 *	11/2004	Barile, Sr.	A47C 3/04
	9,380,870 B2	7/2016	Babick				297/448.1
	9,399,876 B2	7/2016	Magnus	2005/0017554 A1	1/2005	Mizelle et al.	
	D764,295 S	8/2016	Gieske et al.	2005/0017558 A1	1/2005	Fewchuk	
	9,420,890 B2	8/2016	Lee	2005/0051575 A1	3/2005	Durivage	
	9,565,949 B2	2/2017	Peterson	2005/0092780 A1	5/2005	Yamana	
	D781,107 S	3/2017	Welsch et al.	2005/0099052 A1	5/2005	Bertolini et al.	
	9,630,539 B2	4/2017	Yokoyama et al.	2005/0127740 A1	6/2005	Dowty	
	D786,575 S	5/2017	Olarte	2005/0140186 A1	6/2005	Piretti	
	9,642,464 B2	5/2017	Magnus	2005/0146180 A1	7/2005	Fisher et al.	
	9,669,972 B2	6/2017	Stratton	2005/0168032 A1	8/2005	Olarte	
	9,693,630 B2	7/2017	Jacobs et al.	2005/0173954 A1	8/2005	Weber et al.	
	9,693,631 B2	7/2017	Jacobs et al.	2005/0194823 A1	9/2005	Perry et al.	
	D804,313 S	12/2017	Rechtes et al.	2005/0194829 A1	9/2005	Aerts	
	D808,676 S	1/2018	Bellissimo	2005/0242651 A1	11/2005	Peitz et al.	
	9,873,354 B2	1/2018	Poulos et al.	2005/0264060 A1	12/2005	Cesaroni et al.	
	D810,573 S	2/2018	Vazin	2006/0037930 A1	2/2006	Darr et al.	
	9,926,104 B2	3/2018	Olarte	2006/0080817 A1	4/2006	Klinker	
	D821,880 S	7/2018	Olarte	2006/0081662 A1	4/2006	Miura	
	10,010,178 B2	7/2018	Maughan	2006/0202536 A1	9/2006	Luchetti et al.	
	10,273,056 B2	4/2019	Hoffman et al.	2006/0238004 A1	10/2006	Conner et al.	
	D847,527 S	5/2019	Borras Pons	2006/0244301 A1	11/2006	Jeffries	
	10,315,800 B2	6/2019	Olarte	2006/0255642 A1	11/2006	Epaud et al.	
	10,349,749 B2	7/2019	Su	2007/0017886 A1	1/2007	Kao	
	10,485,348 B1	11/2019	Olarte	2007/0018495 A1 *	1/2007	Xiang	A47C 31/02
	10,517,400 B2 *	12/2019	Olarte				297/452.59
	10,518,933 B2	12/2019	Dorn	2007/0029858 A1	2/2007	Grable et al.	
	10,555,610 B2	2/2020	Jacobs et al.	2007/0040090 A1	2/2007	Fay	
	D876,852 S	3/2020	Bellissimo	2007/0063568 A1	3/2007	Figueras Mitjans	
	10,588,414 B2	3/2020	Olarte	2007/0120402 A1	5/2007	Bonzini	
	10,647,471 B2	5/2020	Olarte	2007/0138849 A1	6/2007	Olarte	
	10,674,825 B2	6/2020	Olarte	2007/0170759 A1	7/2007	Nolan et al.	
	10,681,983 B2	6/2020	Olarte	2007/0210638 A1	9/2007	Adragna et al.	
	10,829,277 B2	11/2020	Olarte	2007/0216205 A1	9/2007	Davis	
	10,835,045 B2	11/2020	Olarte	2008/0036266 A1	2/2008	Batley et al.	
	D906,118 S	12/2020	Olarte	2008/0084104 A1	4/2008	VanHorn	
	10,881,210 B2	1/2021	Olarte	2008/0105173 A1	5/2008	Weber	
	10,944,890 B2	3/2021	Tsuchiya et al.	2008/0203790 A1	8/2008	Olarte	
	10,994,890 B2	5/2021	Olarte	2008/0284215 A1	11/2008	Stauber	
	11,008,108 B2	5/2021	Gilbert	2009/0039078 A1	2/2009	Sanfilippo et al.	
	11,026,515 B2	6/2021	Olarte	2009/0045226 A1	2/2009	Munlin	
	11,071,389 B2	7/2021	Olarte	2009/0051207 A1	2/2009	Behrens et al.	
	11,224,292 B2	1/2022	Hashimoto et al.	2009/0121534 A1	5/2009	Brauning	
	11,457,744 B2	10/2022	Olarte	2009/0139054 A1	6/2009	Burnley	
	11,464,337 B1	10/2022	Olarte	2009/0255893 A1	10/2009	Zummo et al.	
	11,470,969 B2	10/2022	Olarte	2009/0273211 A1	11/2009	Hancock et al.	
	11,505,367 B2	11/2022	Olarte	2009/0284057 A1 *	11/2009	Kulish	A47C 3/04
	D979,272 S	2/2023	Olarte				297/239
	11,684,160 B1	6/2023	Olarte	2010/0171345 A1	7/2010	Powell et al.	
	11,690,458 B2 *	7/2023	Politicchio	2010/0176646 A1	7/2010	Rowland et al.	
				2010/0180399 A1	7/2010	Patzler et al.	
				2010/0194160 A1	8/2010	Machael et al.	
				2010/0201165 A1	8/2010	Dankovich	
				2010/0289306 A1 *	11/2010	Saul	A47C 3/04
							297/239
	D1,019,161 S	3/2024	Olarte	2011/0029123 A1	2/2011	Olarte	
	12,011,099 B2	6/2024	Olarte	2011/0108691 A1	5/2011	Alves et al.	
	12,064,035 B2	8/2024	Olarte	2011/0115269 A1	5/2011	Roeder et al.	
	12,150,558 B2	11/2024	Olarte	2011/0198902 A1	8/2011	Renbarger et al.	
	2001/0023510 A1	9/2001	Masubuchi	2011/0233236 A1	9/2011	Brown et al.	

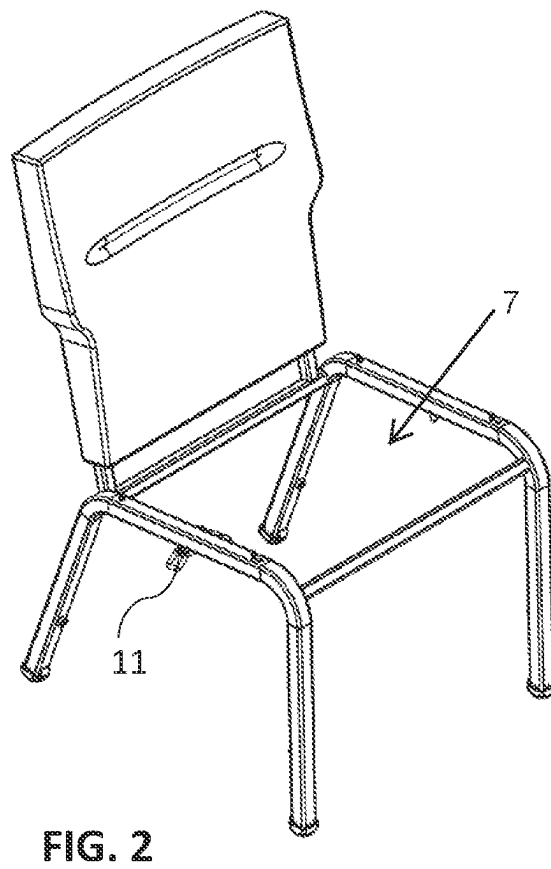
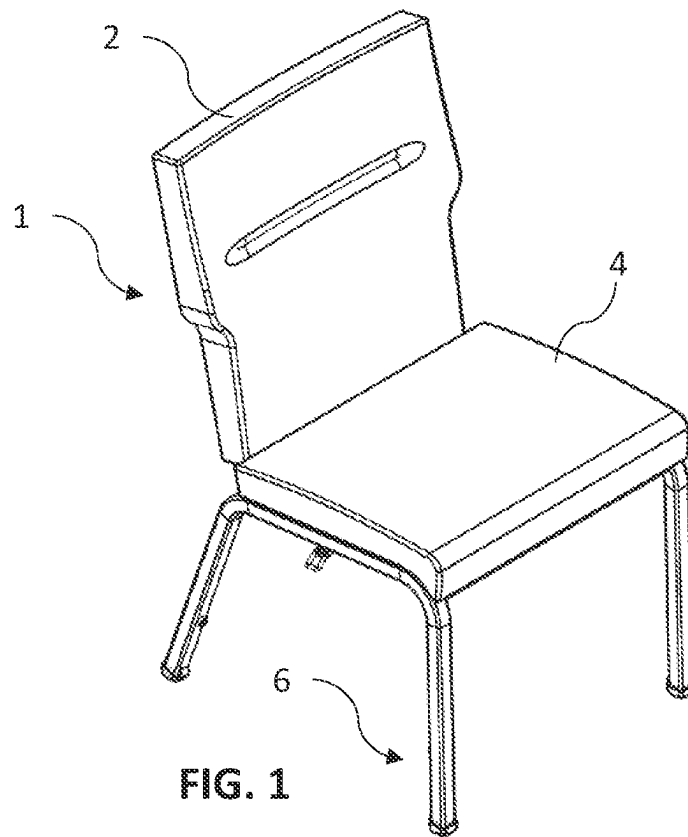
(56)

References Cited

FOREIGN PATENT DOCUMENTS

U.S. PATENT DOCUMENTS							
2012/0126596	A1 *	5/2012	Olarte	A47C 3/04 297/239	AU	2019204079	B2 1/2020
2012/0193318	A1	8/2012	Meager		CA	2465896	5/2003
2012/0228338	A1	9/2012	Magley, II		CA	2465896	A1 5/2003
2012/0228918	A1	9/2012	Takahashi et al.		CA	2678854	8/2008
2012/0247313	A1	10/2012	Peters		CA	2678854	A1 8/2008
2012/0248837	A1	10/2012	Peters		CA	2847538	3/2013
2012/0261971	A1	10/2012	Olarte		CA	2847538	A1 3/2013
2012/0280001	A1	11/2012	Leadbeater		CA	2904885	10/2014
2012/0325718	A1	12/2012	Larsen		CA	2904885	A1 10/2014
2013/0057036	A1	3/2013	Olarte		CA	3045820	12/2019
2013/0082500	A1	4/2013	Line et al.		CA	3045820	A1 12/2019
2013/0255797	A1	10/2013	Coulon et al.		CA	3045828	12/2019
2013/0299522	A1	11/2013	Magley, II		CA	3045828	A1 12/2019
2014/0021222	A1	1/2014	Forbis et al.		CA	3116366	5/2020
2014/0144948	A1	5/2014	Brausen		CA	3116366	A1 5/2020
2014/0165331	A1	6/2014	Kang et al.		CA	3152059	9/2022
2014/0225404	A1	8/2014	Ramirez Magaña		CA	3152059	A1 9/2022
2014/0232153	A1	8/2014	Bell et al.		CN	202908255	5/2013
2014/0239678	A1	8/2014	Todd		CN	208610259	3/2019
2014/0252836	A1	9/2014	Olarte		DE	1821874	11/1960
2014/0283337	A1	9/2014	Triebold et al.		DE	19530156	2/1997
2014/0325920	A1	11/2014	Ugolini et al.		DE	102007024550	11/2008
2014/0332568	A1	11/2014	Stratton		EP	0188002	7/1986
2015/0015051	A1 *	1/2015	Huang	A47C 3/04 297/445.1	EP	0241628	A1 * 10/1987
2015/0021964	A1	1/2015	Jacobs et al.		EP	2114199	9/2012
2015/0028641	A1	1/2015	Stent et al.		EP	2114199	B1 9/2012
2015/0054322	A1	2/2015	Jacobs et al.		FR	2167714	8/1973
2015/0130236	A1	5/2015	Rogers		FR	2167714	8/1973
2015/0150380	A1	6/2015	Oh et al.		GB	180054	5/1922
2015/0150381	A1	6/2015	Lee		GB	454332	9/1936
2015/0158213	A1	6/2015	Mogi et al.		GB	493316	10/1938
2015/0282622	A1	10/2015	Kumazawa		GB	637911	5/1950
2015/0296986	A1	10/2015	Rogers		GB	1133972	11/1968
2015/0374130	A1	12/2015	Jacobs et al.		GB	1464502	2/1977
2016/0255960	A1 *	9/2016	Piretti	A47C 7/002	GB	2098471	11/1982
2016/0288910	A1	10/2016	Udriste et al.		GB	2274391	7/1994
2016/0316916	A1 *	11/2016	Pectol	A47C 7/445	GB	2343443	5/2000
2016/0331138	A1	11/2016	Olarte		JP	01175564	7/1989
2016/0331143	A1	11/2016	Johnson et al.		JP	06198087	7/1994
2017/0029168	A1	2/2017	Olarte		JP	H1191754	4/1999
2017/0057130	A1	3/2017	Sameshima et al.		JP	2005527438	9/2005
2017/0079440	A1	3/2017	Todd		JP	2012249948	12/2012
2017/0127836	A1 *	5/2017	Maughan	A47C 3/04	KR	200341044	2/2004
2017/0164746	A1	6/2017	Phillips		WO	WO9608985	3/1995
2017/0297766	A1	10/2017	Olarte		WO	WO0016661	3/2000
2017/0320415	A1	11/2017	Mueller et al.		WO	2003039297	A2 5/2003
2017/0327281	A1	11/2017	Cross		WO	2005069799	A2 8/2005
2018/0098634	A1	4/2018	Smith et al.		WO	WO2005069799	8/2005
2018/0177300	A1	6/2018	Jacobs et al.		WO	WO2007034007	3/2007
2019/0176669	A1	6/2019	Noro et al.		WO	2008103323	A1 8/2008
2019/0281973	A1	9/2019	Coates et al.		WO	WO2008103323	8/2008
2019/0329673	A1	10/2019	Popescu		WO	WO2011022585	2/2011
2019/0374035	A1	12/2019	Olarte		WO	WO2012054799	4/2012
2020/0154892	A1	5/2020	Olarte		WO	2012142394	A1 10/2012
2020/0189739	A1	6/2020	Schultz et al.		WO	WO2012142394	10/2012
2020/0216178	A1	7/2020	Lin et al.		WO	2013036514	A1 3/2013
2021/0052077	A1	2/2021	Smith et al.		WO	WO2013036514	3/2013
2021/0219728	A1	7/2021	Plant et al.		WO	WO2013172529	11/2013
2021/0321776	A1	10/2021	Plant et al.		WO	2014164246	A1 10/2014
2021/0393035	A1 *	12/2021	Arnold	A47C 4/03	WO	2014164246	10/2014
2022/0047082	A1 *	2/2022	Andersson	F16B 12/44	WO	2016183299	A1 11/2016
2022/0361677	A1	11/2022	Olarte		WO	WO2016183299	11/2016
2023/0047482	A1	2/2023	Olarte		WO	WO2017155562	9/2017
2023/0312107	A1	10/2023	Valdes de la Garza		WO	WO2019065589	4/2019
2023/0320489	A1	10/2023	Olarte		WO	2020102690	A1 5/2020
					WO	WO2020102690	5/2020

* cited by examiner



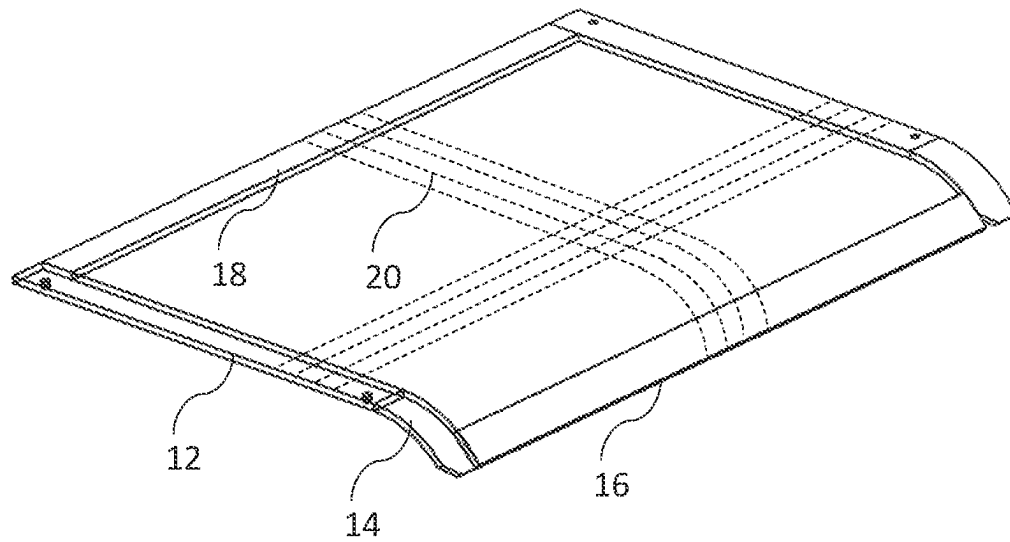


FIG. 3

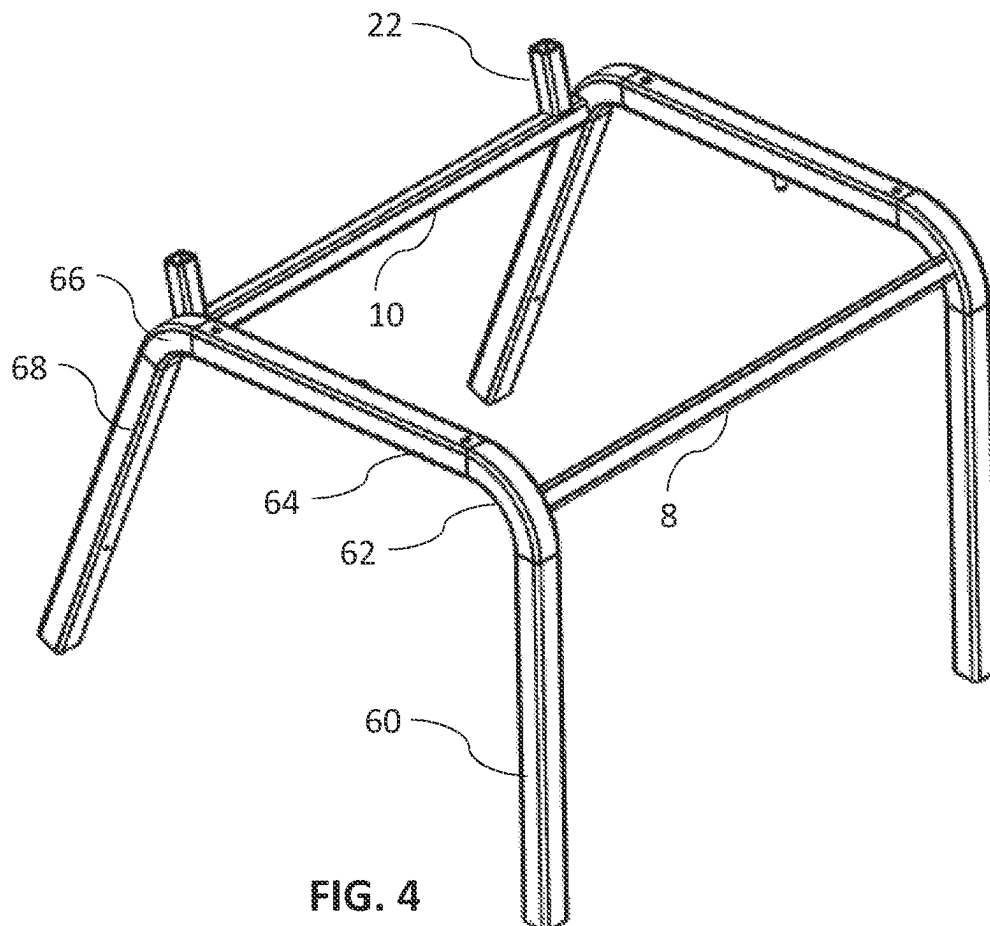


FIG. 4

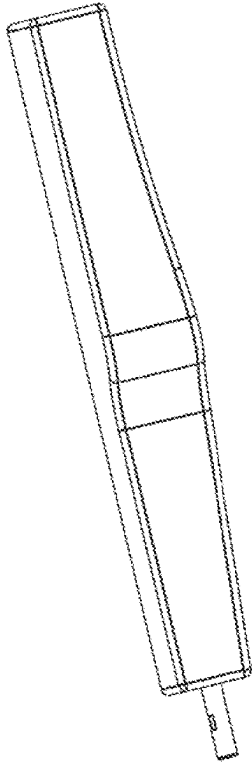


FIG. 5

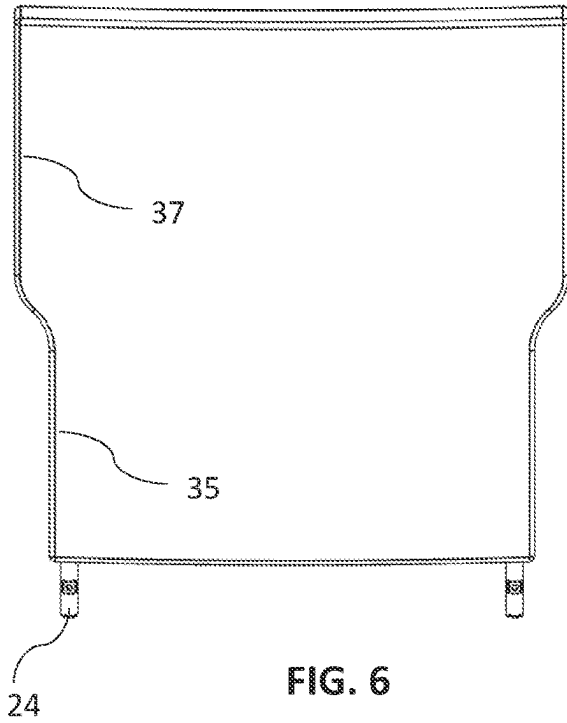


FIG. 6

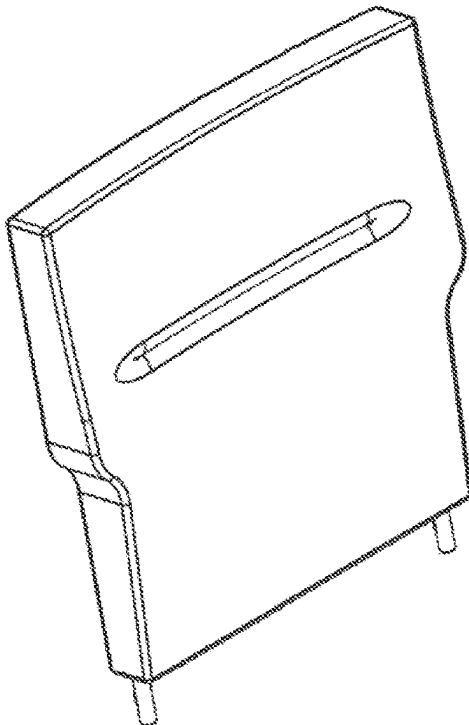


FIG. 7

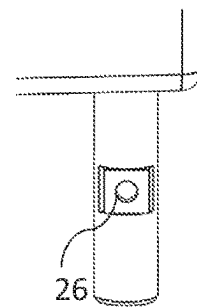


FIG. 8

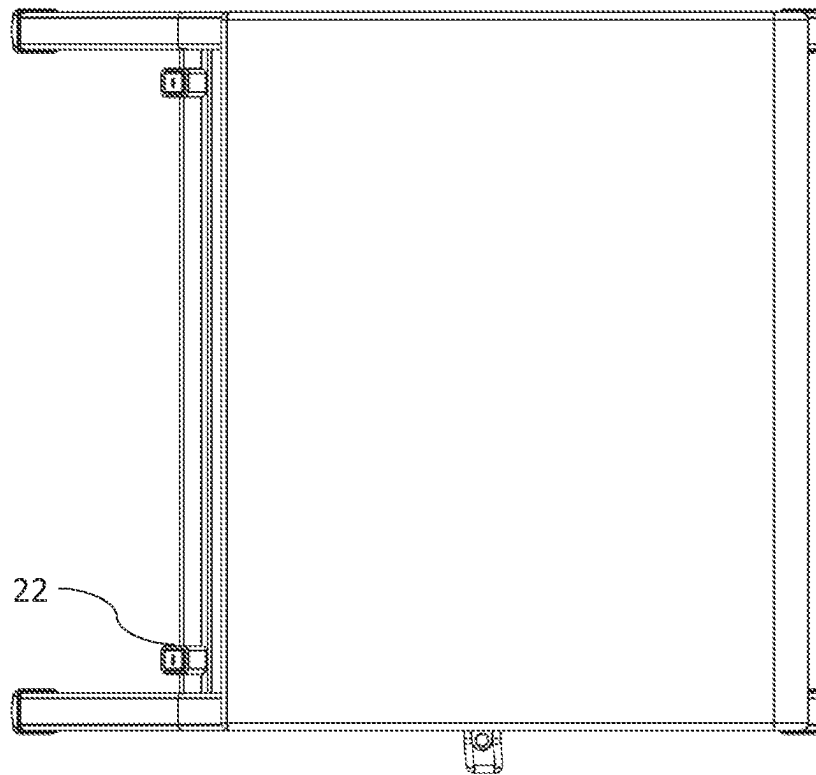


FIG. 9

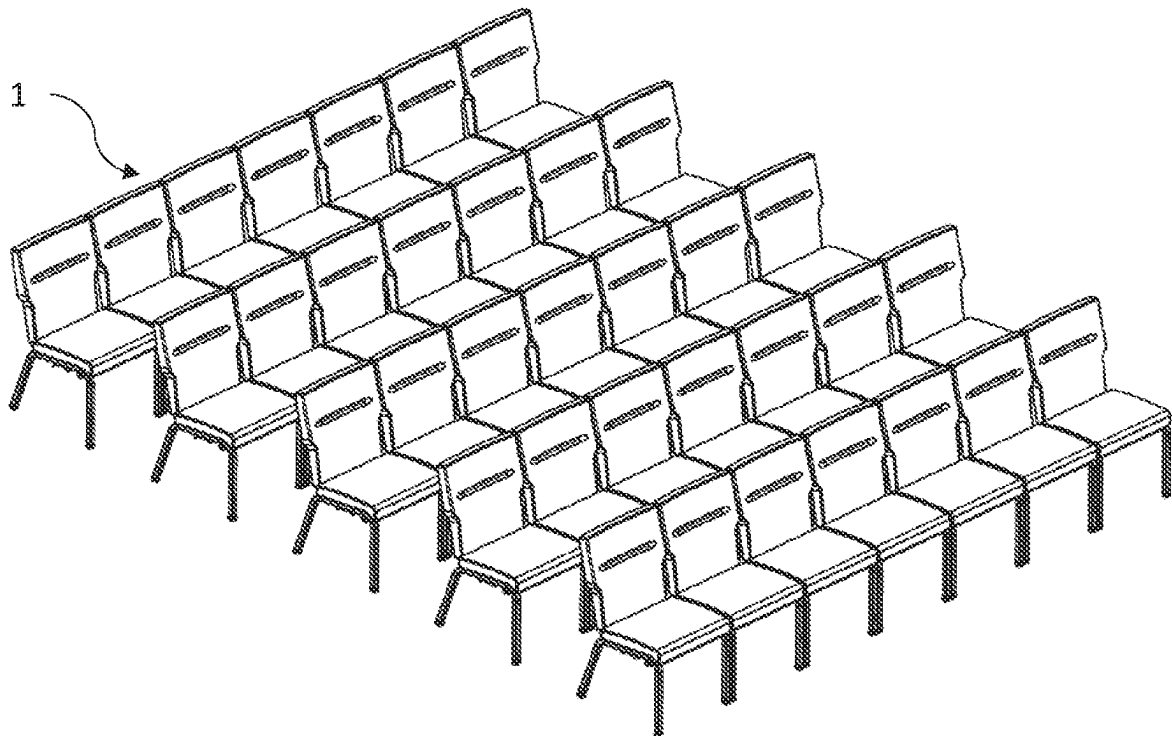


FIG. 10

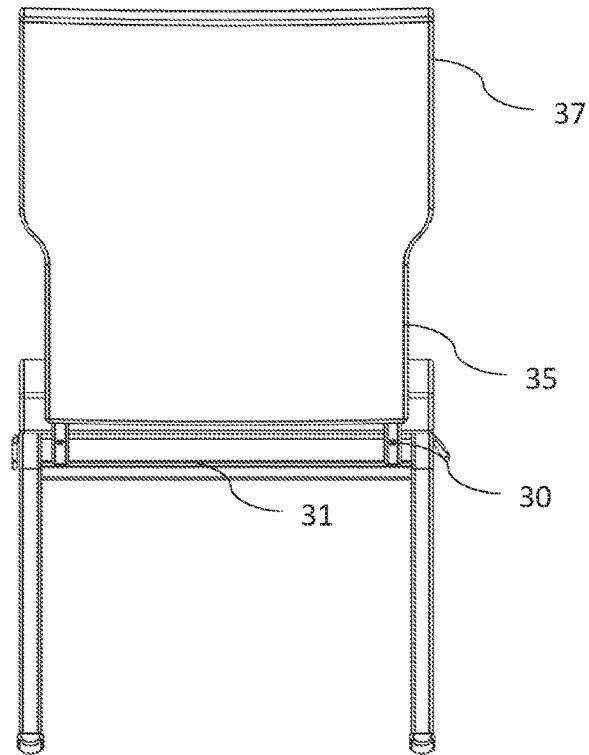


FIG. 11

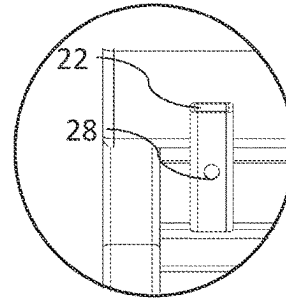


FIG. 12

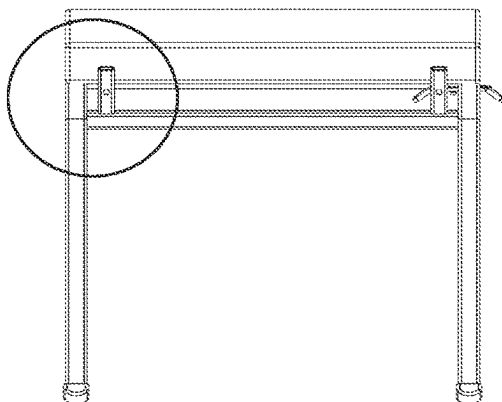


FIG. 13

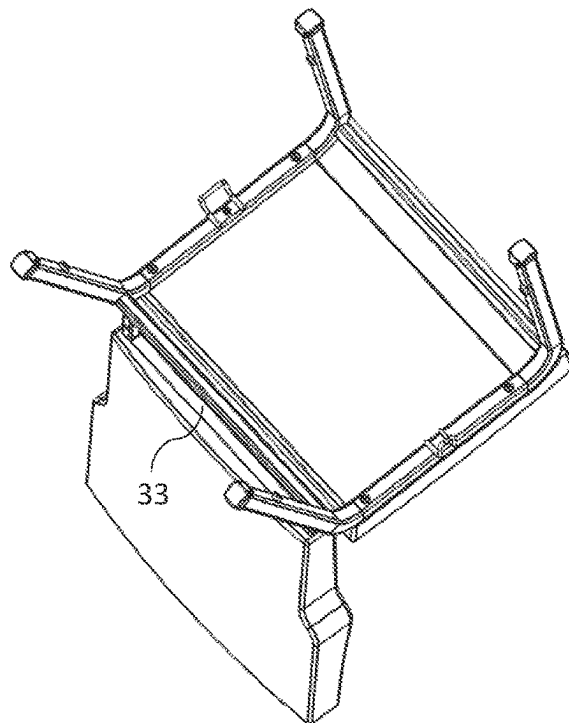


FIG. 14

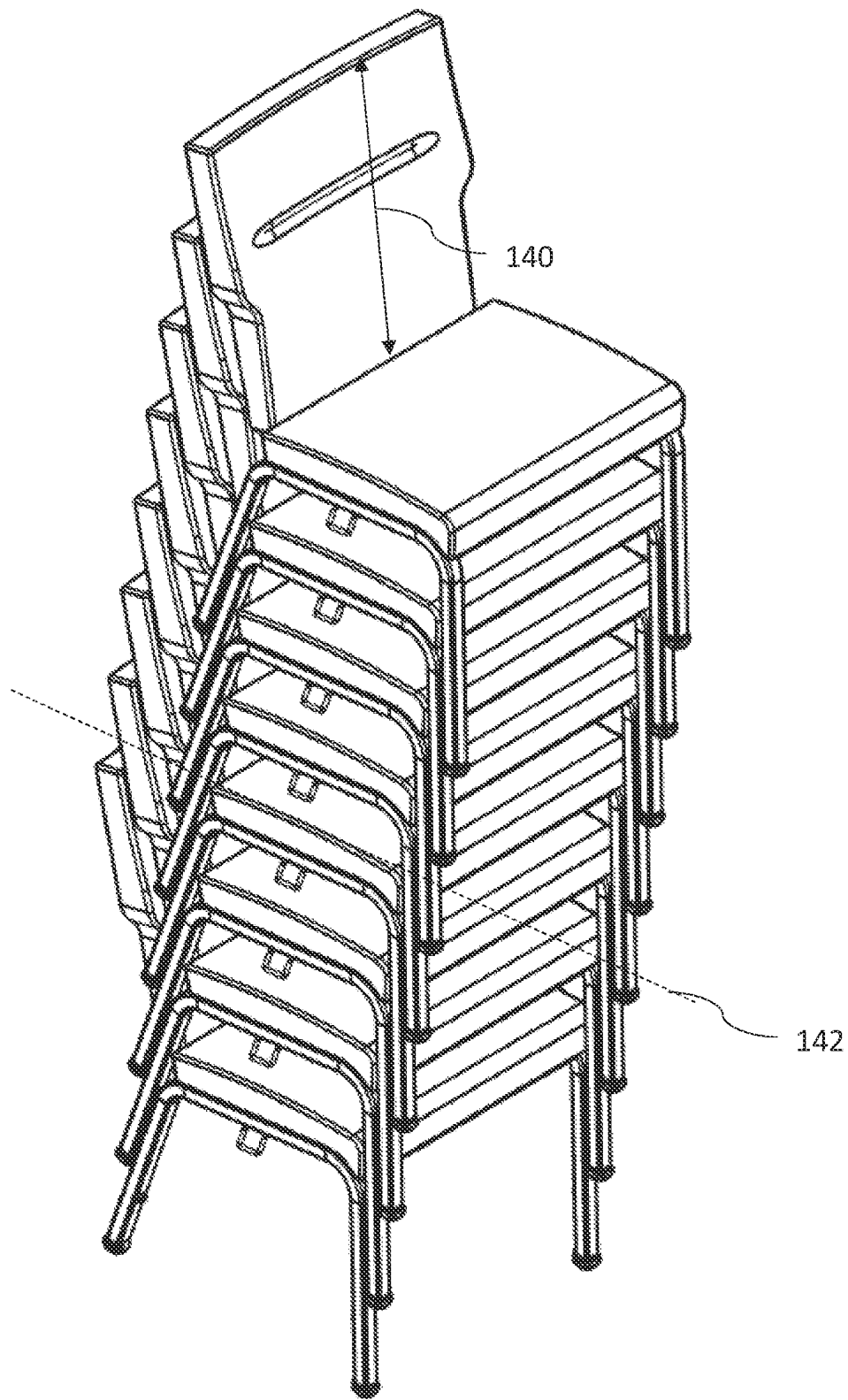


FIG. 15

1

STACKING CHAIR WITH REMOVABLE BACK

FIELD OF THE INVENTION

The following relates to chairs and more particularly moveable chairs which can be used when needed and taken out of the way when not.

BACKGROUND OF THE INVENTION

Stacking chairs are used in event spaces that require flexibility. These chairs will be placed on carts when not in use and put into storage. When needed for an event, the chairs can be arranged in rows for e.g. a conference/convention or church and then moved out of the way when a more open space plan is needed. As a result of these chairs being moved much more frequently than chairs which are bolted to the floor (fixed chairs), the upholstery on these moveable chairs can be come damaged and dirty more frequently.

Furthermore, some known designs of moveable chairs can be very expensive to transport in large quantities due to their construction.

SUMMARY OF THE INVENTION

It is therefore an object of the present invention to provide a comfortable stacking chair which allows for easy replacement of upholstery.

It is a further object of the present invention to provide a sturdy an economical stacking chair which can be easily broken down for shipping to conserve space.

These and other objects are achieved by providing a stacking chair whose back can be removed and seat removed to allow for both easy shipping and easy removal and replacement of the upholstery.

In certain aspects the stacking chair system is provided comprising a frame with four legs, two being front legs and two being rear legs. The seat is secured to the frame in a fixed position, the seat being upholstered and including a foam cushion supported by flexible support. Two receivers are provided in the frame at a rear of the frame. The backrest includes a foam cushion which is upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. A fastener is provided for each receiver and is configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.

In certain aspects the frame includes a support bar extending on either side of the seat and connected to a front curved section and a rear curved section, the front curved section connected to one of the front legs and the rear curved section connected to one of the rear legs, each of the four legs extending outwards and downwards with respect to the seat and the four legs in a fixed arrangement with respect to the seat. In other aspects the two front legs are arranged at a more vertical angle than the rear legs. In other aspects the seat includes rigid supports arranged around a peripheral edge thereof to define a rigid frame and wherein the flexible support is secured to the rigid frame and the rigid frame is removably secured to a side section of the frame. In still other aspects the foam cushion is arranged on top of the rigid frame.

2

In certain aspects the frame includes a rear cross bar and the each of two receivers are arranged as a tube extending from the cross bar, the two receivers spaced apart at a distance which is at least 50% or more a width of the seat, but less than a spacing between the rear legs. In still other aspects the seat includes a seat frame and the flexible support is secured to the seat frame, the seat frame includes a curved portion such that the seat includes a lower curved section which matches a curve of the frame adjacent the front legs.

In yet other aspects the seat frame is secured to the frame such that the seat frame rests on top of the frame with upholstery between the seat and the seat frame and fasteners passing through the frame and securing to the seat frame. In yet other aspects the receiver defines an opening with four flat sides. In still other aspects the protrusion includes a circular cross section. In other aspects the protrusion includes a boss defining a flat section and having a threaded feature adjacent thereto for receiving the fasteners. In other aspects the frame includes a rear cross bar and the receivers are secured to the rear cross bar at an attachment point which is lower than a lower surface of a flat section of the frame which is perpendicular to the rear cross bar.

In certain aspects a stacking chair system is provided including a frame with four legs, two being front legs and two being rear legs. A seat is secured to the frame in a fixed position, the seat including a seat frame which defines a perimeter with an interior opening and the seat frame includes at least a flat portion and a curved portion, the curved portion having a curve which matches a curve of the frame ad or adjacent the front legs and the flat portion configured to rest on top of flat portions of the frame, the seat frame contained in upholstery and the seat including a foam cushion positioned above the seat frame wherein the foam cushion is supported by a flexible support, the flexible support connected to the seat frame and extending across the interior opening. In other aspects seat fasteners pass through holes in the flat portions of the frame and through the upholstery to secure to threaded members of the seat frame. In still other aspects at least part of the upholstery is positioned between the seat frame and the flat portions of the frame with the upholstery in contact with the flat portions of the frame. In yet other aspects two receivers are provided in the frame at a rear of the frame and between the rear legs of the frame and the backrest includes a foam cushion which is upholstered and an inner frame supporting the foam cushion. The inner frame includes two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. In still other aspects, a fastener is provided for each receiver configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners. In yet other aspects the frame defines an opening which is a through opening and at least part of the seat is arranged to move into said opening when a user sits on the seat. In still other aspects the frame defines an opening which is a through opening and at least part of the seat is arranged to move through said opening when a user sits on the seat. In yet other aspects the opening is defined by at least four straight sections of the frame.

Other objects of the invention and its particular features and advantages will become more apparent from consideration of the following drawings and accompanying detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a chair according to the present invention.

3

FIG. 2 is a perspective view of the chair of FIG. 1 with certain components removed.

FIG. 3 is a perspective view of a component of the chair of FIG. 1.

FIG. 4 is a perspective view of the bottom frame section of the chair of FIG. 1.

FIG. 5 is a side view of the backrest of the chair of FIG. 1.

FIG. 6 is a rear view of the backrest of FIG. 5.

FIG. 7 is a perspective view of the backrest of FIG. 5.

FIG. 8 is a detail view of part of FIG. 6.

FIG. 9 is a top view of the chair of FIG. 1 with the backrest removed.

FIG. 10 shows a perspective view of multiple chairs according to FIG. 1 arranged in rows.

FIG. 11 is a rear view of the chair of FIG. 1.

FIG. 12 is a detail view of FIG. 13.

FIG. 13 is a rear view of the chair of FIG. 1 with the backrest removed.

FIG. 14 is a bottom perspective view the chair of FIG. 1.

FIG. 15 is a perspective view of multiple chairs according to FIG. 1 stacked on top of each other.

DETAILED DESCRIPTION OF THE INVENTION

Referring now to the drawings, wherein like reference numerals designate corresponding structure throughout the views. The following examples are presented to further illustrate and explain the present invention and should not be taken as limiting in any regard.

The chair 1 includes a backrest 2 and a seat 4. The seat 4 sits on top of frame 6 and the seat 4 is in a fixed position relative to the frame as is the backrest in that the seat 4 and backrest 2 do not have a tilting function. Connector piece 11 can be provided to assist in holding chairs together/in place in rows (FIG. 10). As can be seen in FIG. 2, the seat 4 can be removed. The backrest 2 and seat 4 are upholstered, thus the removal of the seat assists in replacing the upholstery cover in the even the originally installed one becomes worn, damaged or dirty. The seat 4 includes a support structure shown in FIG. 3 which is positioned under the foam cushion and on top of the lower frame (FIG. 4). The cushion support frame pieces Dec. 14, 2016/18 are preferably welded together and a flexible support 20 is placed over/around these pieces. On the seat frame, piece 14 is curved to match the curve of the corresponding frame section 62. The seat frame pieces Dec. 14, 2016/18 are preferably made from a relatively thin material, for example 1/2" or thinner metal, more preferably 3/8" or 1/4" or thinner metal such as steel. In preferred embodiments, the metal includes threaded holes to receive the screws that secure the seat 4 to the frame 6 from underneath.

Due to the use of the flexible support 20 and the foam placed thereon or molded there around, the bottom of the seat 4 will displace into or through the opening 7 when the seat is weighted, for example when someone sits on the seat. This flexible support 20 may be webbing or a seat sling material as example, but other materials may be used. A more rigid material may also be used for the support 20. A piece of foam rests on top of the flexible support 20 and the upholstery cover is placed over that assembly. The support frame can be attached/removed from the lower frame (FIG. 4) with screws. In this way, the screws can be removed and the seat frame and cushion pulled off the chair. The upholstery cover shown on the seat 4 is preferably removable, for example with a zipper, hook and loop fasteners (e.g. Velcro)

4

or the like using a flap whose seam is shown in FIG. 11 (31). This allows for replacement/cleaning/repair of the upholstery cover as needed. Often chairs of the design shown are used in places of worship and thus the ability for user replacement of the upholstery is an important value feature. Specifically, places of worship would be able to gather volunteers to do a project of replace/renew all of the upholstery on the stacking chairs. As a result, the labor can be donated by members and the cost of having near new chairs is only the cost of new upholstery covers. Thus, the ease of re-upholstery for fixed seat stacking chairs is a significant advantage. Additionally, commercial venues can also replace upholstery with relative ease and a modest labor expense.

FIG. 4 shows additional details on the lower frame construction. Leg 60 is connected to curved section 62 which is connected to horizontal section 64. Another curved section is connected between horizontal section 64 and the rear leg 68. All of these sections 60/62/64/68 of the frame are fixed relative to each other, e.g. by welding or the sections can be formed by bending square tubing. Cross bars 10/8 connect each side of the frame and cross bar 10 includes back supports 22, preferably a welded connection is used. Preferably the frame pieces are hollow and particularly the back support 22 is hollow. Generally the square tubing will have chamfered or rounded corners. As shown in FIGS. 5-8, the back includes lower extensions 24 or protrusions which protrude from below the backrest's upholstered cushion. Like the seat upholstery, the back upholstery also has removable fasteners to enable removal and replacement (or cleaning) of the upholstery cover. FIG. 14 shows the flap 33 which can be opened to allow for removal of the upholstery cover. The lower extension 24 is shown as a round tube and boss 26 provides a flat surface to interface with the interior square tube of the back supports 22. The box 26 also includes a threaded hole, allowing screws to secure the backrest to the lower frame. Further, the lower section of the back 35 is narrower than the upper section 37 and there are curved and/or angled sections between the narrower 35 and wider 37 sections.

As shown in FIGS. 9, 11 and 12, the back supports 22 are hollow and include a hole 28 through which a screw 30 can pass to secure the backrest 2 to the lower frame 6. These screws can also be removed to allow the backrest to likewise be removed. The preferred configuration when shipping chairs is with the backrest removed. Shipping typically has constrained height spaces in that the stack of chairs will cause interference with the backrest and the ceiling of the container/truck in which the chairs are placed. This space 140 is not well used in that the height of the backrest would allow about two or three additional chairs to be accommodated in that otherwise empty vertical space. This can be seen in the stack at FIG. 15 in that the top of the back of the lowest chair is roughly at the level of the seat of the 4th chair in the stack 142, certainly two lower frames with upholstery and no backrest can fit in this space 140. Then, the backrests can be nested together in a stack with maximum density in that the front surface of one back is in contact with a rear surface of another and so on. Thus, there is minimum empty space in the container/truck, and specifically as to the backrests. As a result, more chairs can be shipped within a given space constraint.

Although the invention has been described with reference to a particular arrangement of parts, features and the like, these are not intended to exhaust all possible arrangements or features, and indeed many other modifications and variations will be ascertainable to those of skill in the art.

5

What is claimed is:

1. A stacking chair comprising:

a frame with four legs, two being front legs and two being rear legs;

a seat secured to the frame in a fixed position, the seat being upholstered and including a foam cushion supported by flexible support;

two receivers in the frame at a rear of the frame;

a backrest comprising a foam cushion which is upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers;

a fastener for each receiver configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.

2. The stacking chair of claim 1 wherein the frame further comprises a support bar extending on either side of the seat and connected to a front curved section and a rear curved section, the front curved section connected to one of the front legs and the rear curved section connected to one of the rear legs, each of the four legs extending outwards and downwards with respect to the seat and the four legs in a fixed arrangement with respect to the seat.

3. The stacking chair of claim 2 wherein the two front legs are arranged at a more vertical angle than the rear legs.

4. The stacking chair of claim 2 wherein the seat includes rigid supports arranged around a peripheral edge thereof to define a rigid frame and wherein the flexible support is secured to the rigid frame and the rigid frame is removably secured to a side section of the frame.

5. The stacking chair of claim 4 wherein the foam cushion of the seat is arranged on top of the rigid frame.

6. The stacking chair of claim 1 wherein the frame further comprises a rear cross bar and the each of two receivers are arranged as a tube extending from the cross bar, the two receivers spaced apart at a distance which is at least 50% or more a width of the seat, but less than a spacing between the rear legs.

7. The stacking chair of claim 1 wherein the seat includes a seat frame and the flexible support is secured to the seat frame, the seat frame includes a curved portion such that the seat includes a lower curved section which matches a curve of the frame adjacent the front legs.

8. The stacking chair of claim 7 wherein the seat frame is secured to the frame such that the seat frame rests on top of the frame with upholstery between the seat and the seat frame and fasteners passing through the frame and securing to the seat frame.

9. The stacking chair of claim 1 wherein the receiver defines an opening with four flat sides.

6

10. The stacking chair of claim 9 wherein the protrusion includes a circular cross section.

11. The stacking chair of claim 10 wherein the protrusion includes a boss defining a flat section and having a threaded feature adjacent thereto for receiving the fasteners.

12. The stacking chair of claim 1 wherein the frame comprises a rear cross bar and the receivers are secured to the rear cross bar at an attachment point which is lower than a lower surface of a flat section of the frame which is perpendicular to the rear cross bar.

13. A stacking chair comprising:

a frame with four legs, two being front legs and two being rear legs;

a seat secured to the frame in a fixed position, the seat including a seat frame which defines a perimeter with an interior opening and the seat frame includes at least a flat portion and a curved portion, the curved portion having a curve which matches a curve of the frame at or adjacent the front legs and the flat portion configured to rest on top of flat portions of the frame, the seat frame contained in upholstery and the seat including a foam cushion positioned above the seat frame wherein the foam cushion is supported by a flexible support, the flexible support connected to the seat frame and extending across the interior opening.

14. The stacking chair of claim 13 further comprising seat fasteners which pass through holes in the flat portions of the frame and through the upholstery to secure to threaded members of the seat frame.

15. The stacking chair of claim 13 wherein at least part of the upholstery is positioned between the seat frame and the flat portions of the frame with the upholstery in contact with the flat portions of the frame.

16. The stacking chair of claim 13 further comprising:

two receivers in the frame at a rear of the frame and between the rear legs of the frame;

a backrest comprising a foam cushion which is upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers;

a fastener for each receiver configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.

17. The stacking chair of claim 13 wherein the frame defines an opening which is a through opening and at least part of the seat is arranged to move into said opening when a user sits on the seat.

18. The stacking chair of claim 17 wherein the opening is defined by at least four straight sections of the frame.

* * * * *